

Lecture 3 redo

- Topics:
- ① camera measurements
 - ② Essential matrix / Triangulation
 - ③ Point / line / Plane
-

The measurements:

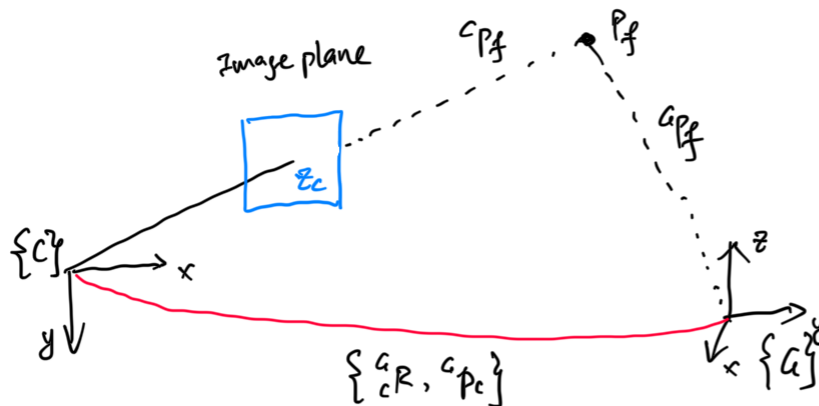
$$z_c = h_c(x) + n_c$$

z_c : pixel measurements

x : state vector: camera pose
point feature

h_c : camera model

n_c : pixel noise



The most simple camera model:

$${}^C P_f = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad z_n = \begin{bmatrix} u_n \\ v_n \end{bmatrix} = \begin{bmatrix} \frac{x}{z} \\ \frac{y}{z} \end{bmatrix}$$

The basic function:

$$z_n = h_p({}^C P_f) + n_c \quad \text{with } {}^C P_f = {}^C R^T ({}^G P_f - {}^G P_C)$$

$$\tilde{z}_n = H_c \tilde{x} + n_c$$

$$\tilde{z}_n = z_n - h_p({}^C \hat{P}_f) = \begin{bmatrix} u_n \\ v_n \end{bmatrix} - \begin{bmatrix} \frac{\hat{x}}{\hat{z}} \\ \frac{\hat{y}}{\hat{z}} \end{bmatrix}$$

$$x: \{ {}^G R, {}^G P_C, {}^G P_f \} \quad {}^G R = {}^G \hat{R} \exp(\delta \theta) \quad {}^G P_C = {}^G \hat{P}_C + {}^G \tilde{P}_C \quad {}^G P_f = {}^G \hat{P}_f + {}^G \tilde{P}_f$$

$$\tilde{x} = \begin{bmatrix} \delta \theta \\ {}^G \tilde{P}_C \\ {}^G \tilde{P}_f \end{bmatrix}$$

$$\tilde{z}_n = H_c \tilde{x} + n_c$$

$$= \begin{bmatrix} \frac{\partial \tilde{z}_n}{\partial \theta} & \frac{\partial \tilde{z}_n}{\partial \hat{p}_f} & \frac{\partial \tilde{z}_n}{\partial \hat{p}_c} \end{bmatrix} \tilde{x} + n_c$$

$$= \begin{bmatrix} \frac{\partial \tilde{z}_n}{\partial \hat{p}_f} \cdot \frac{\partial \hat{p}_f}{\partial \theta} & \frac{\partial \tilde{z}_n}{\partial \hat{p}_f} \cdot \frac{\partial \hat{p}_f}{\partial \hat{p}_f} & \frac{\partial \tilde{z}_n}{\partial \hat{p}_f} \cdot \frac{\partial \hat{p}_f}{\partial \hat{p}_c} \end{bmatrix} \tilde{x} + n_c$$

$$= \frac{\partial \tilde{z}_n}{\partial \hat{p}_f} \begin{bmatrix} \frac{\partial \hat{p}_f}{\partial \theta} & \frac{\partial \hat{p}_f}{\partial \hat{p}_f} & \frac{\partial \hat{p}_f}{\partial \hat{p}_c} \end{bmatrix} \tilde{x} + n_c$$

$$\tilde{z}_n = \begin{bmatrix} \frac{c_x}{c_z} \\ \frac{c_y}{c_z} \end{bmatrix} \quad \frac{\partial \tilde{z}_n}{\partial \hat{p}_f} = \begin{bmatrix} \frac{1}{c_z} & 0 & -\frac{c_x}{c_z^2} \\ 0 & \frac{1}{c_z} & -\frac{c_y}{c_z^2} \end{bmatrix} = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix}$$

$$\hat{p}_f = \hat{c}R^T(\hat{q}_f - \hat{q}_c)$$

$$\Rightarrow \hat{q}_f + \hat{p}_f = (\hat{c}R \exp(\delta\theta))^T (\hat{q}_f + \hat{q}_f - \hat{q}_c - \hat{q}_c)$$

$$= (I - \delta\theta I_x) \hat{c}R^T (\hat{q}_f + \hat{q}_f - \hat{q}_c - \hat{q}_c)$$

$$\approx \hat{c}R^T (\hat{q}_f - \hat{q}_c) - \delta\theta I_x \hat{c}R^T (\hat{q}_f - \hat{q}_c) + \hat{c}R^T (\hat{q}_f - \hat{q}_c)$$

$$\hat{q}_f + \hat{p}_f = \hat{c}R^T (\hat{q}_f - \hat{q}_c) + \delta\theta \hat{c}R^T (\hat{q}_f - \hat{q}_c) - \hat{c}R^T \hat{q}_c + \hat{c}R^T \hat{q}_f$$

$$\hat{p}_f = \delta\theta \hat{c}R^T (\hat{q}_f - \hat{q}_c) - \hat{c}R^T \hat{q}_c + \hat{c}R^T \hat{q}_f$$

$$\hat{q}_f = \hat{c}R^T (\hat{q}_f - \hat{q}_c)$$

$$\frac{\partial \hat{p}_f}{\partial \theta} = \hat{c}R^T I_x \quad \frac{\partial \hat{p}_f}{\partial \hat{p}_c} = -\hat{c}R^T \quad \frac{\partial \hat{p}_f}{\partial \hat{p}_f} = \hat{c}R^T$$

$$\text{then: } \tilde{z}_n = H_c \tilde{x} + n$$

$$\tilde{z}_n = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix} \begin{bmatrix} \hat{c}R^T I_x & -\hat{c}R^T & \hat{c}R^T \end{bmatrix} \tilde{x} + n_c$$

A few important discussions:

- ① SLAM:
 $\{ {}^G C_R, {}^G C_P \}$ is the pose, localization of camera
 $\{ C_{Pf} \}$ is the 3D feature of the environment

Solving them together means we solve localization and mapping together $\Rightarrow$ Simultaneous localization and mapping (SLAM)

- ② If C_{Pf} is known, given (u, v) , how to solve $\{ {}^G C_R, {}^G C_P \}$?

In this case, $x: \{ {}^G C_R, {}^G C_P \}$, 6 dof. Z_n measurements is 2 dof

$$Z_n = h(\underbrace{{}^G C_R, {}^G C_P}_{6 \text{ dof unknown}}) + n_c$$

we need at least 3 points: $C_{Pi} \leftrightarrow (u_i, v_i)$ to solve $\{ {}^G C_R, {}^G C_P \}$.

If use 3 pairs $C_{Pi} \leftrightarrow (u_i, v_i)$, P3P: perspective-3-points: 3+1 points
 use 3D $\leftrightarrow$ 2D correspondences to determine the same 3D pose

If use n pairs of 3D $\leftrightarrow$ 2D correspondences, PnP: perspective- n -points.

RANSAC: random sample consensus for outliers rejection. structure $\rightarrow$ motions

- ③ If: $\{ {}^G C_R, {}^G C_{P2} \}$ is known.

given: $(u_1, v_1), (u_2, v_2) \rightarrow P_f?$

$$(u_1, v_1) \rightarrow {}^C_1 b_1 = \frac{\begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix}}{\| \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} \|}, \quad r_1: \text{range}$$

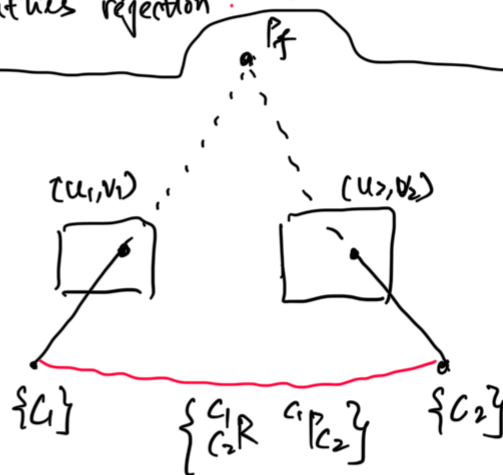
$$(u_2, v_2) \rightarrow {}^C_2 b_2 = \frac{\begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix}}{\| \begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix} \|}, \quad r_2: \text{range}$$

$${}^C_1 P_f = {}^C_1 b_1 \cdot r_1, \quad {}^C_2 P_f = {}^C_2 b_2 \cdot r_2$$

r_1, r_2 is unknown.

$${}^C_1 P_f = {}^C_2 R {}^C_2 P_f + {}^C_1 P_{C2} \Rightarrow {}^C_1 b_1 \cdot r_1 = {}^C_2 R \cdot {}^C_2 b_2 \cdot r_2 + {}^C_1 P_{C2}$$

way 1:
 $\Rightarrow {}^C_1 b_1 r_1 - {}^C_2 R {}^C_2 b_2 r_2 = {}^C_1 P_{C2} \Rightarrow \begin{bmatrix} {}^C_1 b_1 & - {}^C_2 R {}^C_2 b_2 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = {}^C_1 P_{C2}$



solve $r_1, r_2 \Rightarrow$ solve ${}^C P_f, {}^C P_g$ 3×2 4×1 2×1

degenerate case: If pure rotation, no translation,

$$\begin{cases} {}^C b = {}^C_2 R {}^C_2 b, & {}^C P_2 = 0_{3 \times 1} \Rightarrow \text{no way to solve } r. \\ r_1 = r_2 \end{cases}$$

for mono cam, pure rotation is dangerous.

way 2: ${}^C b \cdot r_1 = {}^C_2 R {}^C_2 b \cdot r_2 + {}^C P_2$ ${}^C b$ we can find: b^{+1}, b^{+2}

$${}^C b^T b^{+1} = 0 \quad {}^C b^T b^{+2} = 0$$

$$[b^{+1} \ b^{+2}]^T {}^C b \cdot r_1 = [b^{+1} \ b^{+2}]^T ({}^C_2 R {}^C_2 b \cdot r_2 + {}^C P_2) = 0_{2 \times 1}$$

$$\Rightarrow \underbrace{[b^{+1} \ b^{+2}]^T {}^C_2 R {}^C_2 b}_{2 \times 1} r_2 = - \underbrace{[b^{+1} \ b^{+2}]^T {}^C P_2}_{2 \times 1}$$

② If $\{ {}^C_2 R \}$ is known, ${}^C P_2$ we only know the direction $\vec{{}^C P_2}$, $\| {}^C P_2 \|$ is unknown

Given $(u_1, v_1), (u_2, v_2)$, can we triangulate ${}^C P_f$?

let: $\| {}^C P_2 \| = p$, then: ${}^C P_f = {}^C_2 R {}^C P_f + {}^C P_2$

$$\Rightarrow {}^C b \cdot r_1 = {}^C_2 R {}^C_2 b \cdot r_2 + p \cdot \vec{{}^C P_2}$$

$$\vec{{}^C P_2}: v^{+1}, v^{+2}$$

$$\vec{{}^C P_2}^T v^{+1} = 0 \quad \vec{{}^C P_2}^T v^{+2} = 0$$

$$A = [v^{+1} \ v^{+2}]^T$$

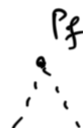
$$A {}^C b \cdot r_1 = A {}^C_2 R {}^C_2 b \cdot r_2 + p \cdot \underbrace{A \vec{{}^C P_2}}_0$$

$$[A {}^C b \quad -A {}^C_2 R {}^C_2 b] \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = 0$$

r_1, r_2 is up to scale. Hence: Mono camera does not have scale.

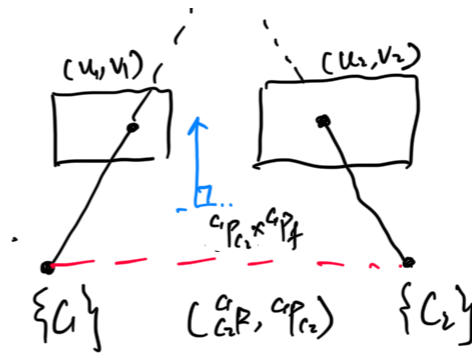
The above is use motion to infer structure

③ Given $\{u_1, v_1\}, \{u_2, v_2\}$



can we $\{C_P, C_P\}$?

$C_{P_f}, C_{P_f}, C_{P_c} \rightarrow \text{triangular}$



$$C_{P_c} \times C_{P_f} \perp C_1^T C_2^T P_f$$

$$(C_{P_c} \times C_{P_f})^T C_1^T C_2^T P_f = 0 \Rightarrow C_{P_f}^T L_{C_{P_c}} \times C_1^T C_2^T P_f = 0$$

$$C_{P_f} = C_b \cdot r_1, C_{P_f} = C_b \cdot r_2$$

$$\Rightarrow C_b^T L_{C_{P_c}} \times C_1^T C_2^T C_b = 0 \quad E = L_{C_{P_c}} \times C_2^T C_1^T \text{ essential matrix.}$$

E is 5 dof? why?

$(u_1, v_1), (u_2, v_2) \rightarrow$ provides 1 constraint

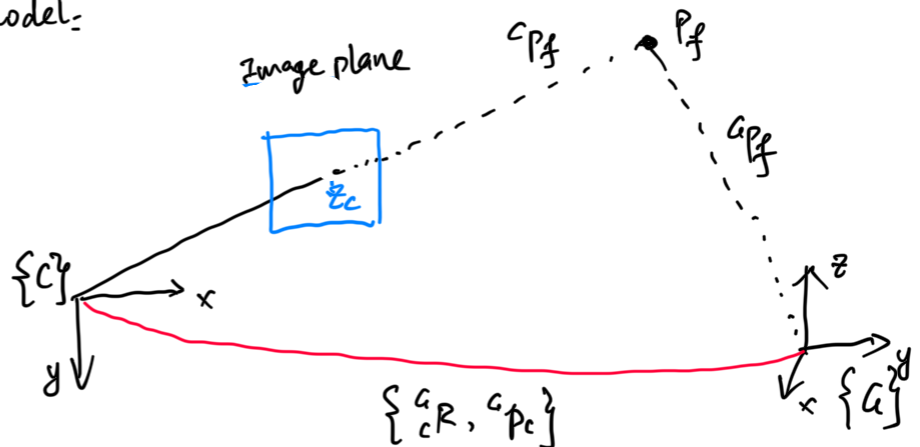
Hence, we need at least 5 pair of points to solve $E \Rightarrow C_1^T C_2^T, C_{P_c}^T$
3dof 2dof

when selecting pair of points, RANSAC comes in:
Random sample consensus

If (u, v) are in normalized coordinates, $C_b^T L_{C_{P_c}} \times C_2^T C_1^T C_b$ Essential matrix
If (u, v) are in pixel coordinates, fundamental matrix.

(motion from structure)

The calibration in camera model:



$$Z_n = \begin{bmatrix} \frac{x_n}{c_z} \\ \frac{y_n}{c_z} \\ 1 \end{bmatrix} \triangleq h_p(p_f) \quad h_p: \text{projection}$$

$$p_f = c_R^T (p_f - p_c) \triangleq h_t(c_R, p_c, p_f) \quad h_t: \text{transform function}$$

$$Z_n = h_p(h_t(c_R, p_c, p_f)) + n$$

Image distortion: $\begin{cases} \text{radical-tangential (for example)} \\ \text{fish eye: equidist model} \end{cases} \quad r^2 = u_n^2 + v_n^2$

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = \begin{bmatrix} u_n \\ v_n \end{bmatrix} (1 + k_1 r^2 + k_2 r^4) + \begin{bmatrix} 2p_1 u_n v_n + p_2 (r^2 + 2u_n^2) \\ p_1 (r^2 + 2v_n^2) + 2p_2 u_n v_n \end{bmatrix}, \quad p_1, k_2, p_1, p_2 \text{ are dist parameters.}$$

$$Z_d = \begin{bmatrix} u_d \\ v_d \end{bmatrix} = h_d(Z_n)$$

$$x_d = [k_1, k_2, p_1, p_2]^T$$

camera intrinsics: homography:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_K \begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix}$$

$$x_{in} = [f_u, f_v, c_u, c_v]$$

$$Z = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} f_u & 0 \\ 0 & f_v \end{bmatrix} \begin{bmatrix} u_d \\ v_d \end{bmatrix} + \begin{bmatrix} c_u \\ c_v \end{bmatrix}$$

$$\triangleq h_k(Z_d, x_{in})$$

Final: $Z_c = h_c(x) + n_c$

$$= h_k(Z_d, x_{in}) + n_c$$

$$= h_k(h_d(Z_n, x_d), x_{in}) + n_c$$

$$= h_k(h_d(h_p(p_f), x_d), x_{in}) + n_c$$

$$= h_k(h_d(h_p(h_t(c_R, p_c, p_f)), x_d), x_{in}) + n_c$$

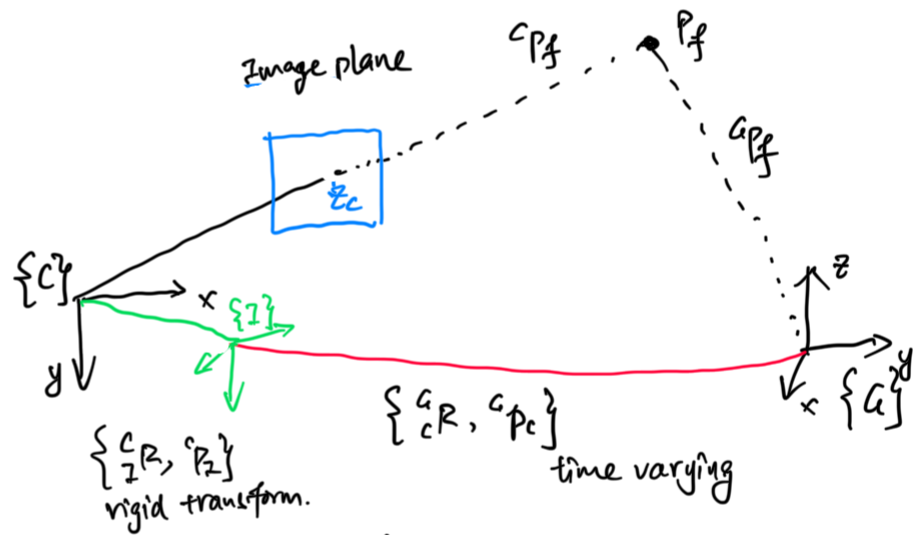
$\left\{ \begin{array}{l} h_c(\cdot): \text{camera intrinsics homography} \\ h_d(\cdot): \text{distortion function} \\ h_p(\cdot): \text{projection function} \\ h_t(\cdot): \text{transform function} \end{array} \right.$

x_d : distortion parameters

x_{in} : intrinsics: f_u, f_v, c_u, c_v

You can use chain rule to compute the Jacobians for calibration

The extrinsic calibration:



camera and IMU sensor frame are rigid connected.

$$\begin{bmatrix} {}^C p_f \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^C R & {}^C P_2 \\ 0_{1 \times 3} & 1 \end{bmatrix}}_{\hat{Z}T} \underbrace{\begin{bmatrix} {}^G R & {}^G P_2 \\ 0_{1 \times 3} & 1 \end{bmatrix}^{-1}}_{\hat{Z}T^{-1}} \begin{bmatrix} {}^G p_f \\ 1 \end{bmatrix}$$

$${}^C p_f = {}^C R {}^G p_f + {}^C P_2 = {}^C R ({}^G R ({}^G p_f - {}^G P_2)) + {}^C P_2$$

Note that: How to estimate the time offset between camera and IMU?

$$t_2 = t_c + t_d$$

$$\begin{bmatrix} {}^{C t_c} p_f \\ 1 \end{bmatrix} = \hat{Z}T \cdot \begin{bmatrix} G \\ I_{(t_c+t_d)} \end{bmatrix}^{-1} \cdot \begin{bmatrix} {}^G p_f \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} G \\ I_{(t_c+t_d)} \end{bmatrix}^{-1} = \begin{bmatrix} {}^G R & {}^G P_2 \\ 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^G w & {}^G v \\ 0_{1 \times 3} & 1 \end{bmatrix} \quad \text{This is the Jacobian for } t_d$$

Then it's the full calibration for vision system.

Point / line / Plane representation:

Point representation:

$$\mathbb{R}^3, \text{ closest point: } P_f = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} = z \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix}$$

Inverse depth:

$$x_f = \begin{bmatrix} x \\ y \\ \frac{1}{z} \end{bmatrix} = \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix} \quad \lambda \triangleq \frac{1}{z}$$

What's the motivation to use inverse depth?

$$\begin{aligned} C_2 P_f &= C_2 R (C_1 P_f - C_1 P_c) \\ &= C_2 R \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - C_1 P_c \right) \\ &= C_2 R \left(z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - C_1 P_c \right) \end{aligned}$$

$$\frac{C_2 P_f}{z} = C_2 R \left(\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - \frac{1}{z} C_1 P_c \right)$$

$$\begin{bmatrix} C_2 x/z \\ C_2 y/z \\ C_2/z \end{bmatrix} = C_2 R \left(\begin{bmatrix} x \cdot \lambda \\ y \cdot \lambda \\ 1 \end{bmatrix} - \lambda C_1 P_c \right)$$

If $z \rightarrow \infty$, the feature is far away, $z \approx C_2 z \Rightarrow C_2 x, C_2 y$,

$$\text{then } \begin{bmatrix} C_2 x/z \\ C_2 y/z \\ C_2/z \end{bmatrix} \approx \begin{bmatrix} u_{n2} \\ v_{n2} \\ 1 \end{bmatrix} \approx C_2 R \left(\begin{bmatrix} x \cdot \lambda \\ y \cdot \lambda \\ 1 \end{bmatrix} - \lambda C_1 P_c \right)$$

measurements
in C_2

$$\frac{\partial \begin{bmatrix} u_{n2} \\ v_{n2} \end{bmatrix}}{\partial \tilde{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} H_0 & H_p & H_{\tilde{x}} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \left[C_2 R \left(\begin{bmatrix} x \cdot \lambda \\ y \cdot \lambda \\ 1 \end{bmatrix} - \lambda C_1 P_c \right) \right]_x$$

$\downarrow$
 $\lambda \rightarrow 0$

$\downarrow$
 0

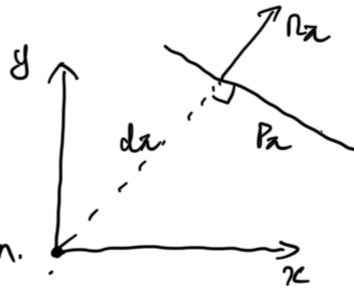
$\downarrow$
 0

If the feature is far away,
then rotation is improved, the Jacobians for translation will not be improved.

Plane representation

$$\pi = \begin{bmatrix} n_x \\ d_x \end{bmatrix}_{4 \times 1}$$

n_x : normal direction
 d_x : distance to origin.

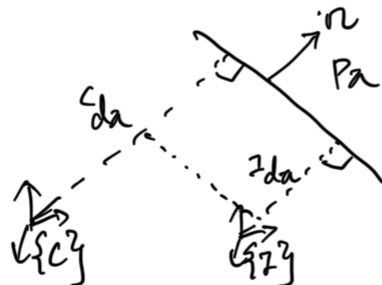


plane is 3dof but π is 4 parameters.

closest point representation:

$$P_x = d_x \cdot n_x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ this is the closest point from plane to the origin.}$$

How to transfer the plane from camera frame to Imu frame?



$${}^c P_x \leftrightarrow {}^I P_x$$

$$\begin{cases} {}^I n_x = {}^I R {}^c n_x & \Rightarrow \text{normal direction} \\ {}^I d_x = {}^c d_x - {}^c n_x^T \cdot {}^c P_I & \Rightarrow \text{origin to plane} \end{cases}$$

$$\text{Hence: } \begin{bmatrix} {}^I n_x \\ {}^I d_x \end{bmatrix} = \begin{bmatrix} {}^I R & 0_{3 \times 1} \\ -{}^c P_I^T & 1 \end{bmatrix} \begin{bmatrix} {}^c n_x \\ {}^c d_x \end{bmatrix}$$

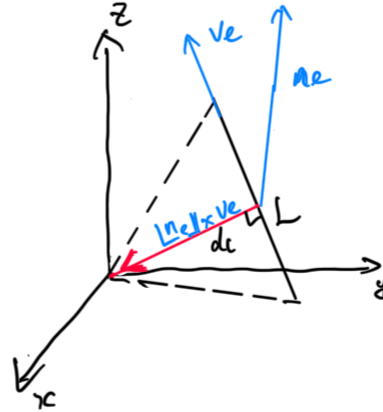
$$\begin{bmatrix} {}^I n_x \\ {}^I d_x \end{bmatrix} = \begin{bmatrix} {}^I R & 0_{3 \times 1} \\ {}^I P_C^T {}^I R & 1 \end{bmatrix} \begin{bmatrix} {}^c n_x \\ {}^c d_x \end{bmatrix}$$

The transformation is as this way.

$${}^C P_A \rightarrow ({}^C n_A, {}^C d_A) \rightarrow ({}^Z R, {}^Z P_C) \rightarrow ({}^Z n_A, {}^Z d_A) \rightarrow {}^Z P_A$$

The line representation

How to represent a 3D line?



For a line: L : v_L is 3d direction
 d_L is distance from origin to the line
 n_L is the normal direction of the plane

Hence: we can represent a line with:

L : n_L, v_L, d_L

$R_L = [n_L \ v_L \ (n_L \times v_L)]$ this is a rotation matrix
 d_L : is a distance scalar.

L : $\{R_L, d_L\} : \{\bar{q}_L, d_L\} : P_L = \bar{q}_L \cdot d_L$ 4D vector:

Error states in Euclidean space

Then: $P_L = \hat{P}_L + \tilde{P}_L$

$$= \bar{q}_L \cdot d_L = \hat{q}_L \otimes \delta q_L \cdot (d_L + \tilde{d}_L)$$

$$\Leftrightarrow \hat{P}_L + \tilde{P}_L = \hat{q}_L \otimes \begin{bmatrix} 1 \\ \frac{1}{2} \delta \theta_L \end{bmatrix} (d_L + \tilde{d}_L)$$

$$= \hat{q}_L \otimes \begin{bmatrix} d_L + \tilde{d}_L \\ \frac{1}{2} d_L \delta \theta_L \end{bmatrix}$$

$$= \hat{q}_L \otimes \left(\begin{bmatrix} d_L \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} d_L \delta \theta_L \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta \theta_L \end{bmatrix} \right)$$

$$\cancel{\hat{P}_L + \tilde{P}_L} = \cancel{\hat{q}_L \otimes} + \hat{q}_L \otimes \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} d_L \delta \theta_L \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta \theta_L \end{bmatrix}$$

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$$\tilde{P}_0 = \begin{bmatrix} \hat{q}_0 & 1 \\ 0 & \hat{q}_0^T \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} d_0 \\ \delta \theta_0 \end{bmatrix}$$

A few discussions on line with cameras:

① How to project 3D lines to 2D line in image?

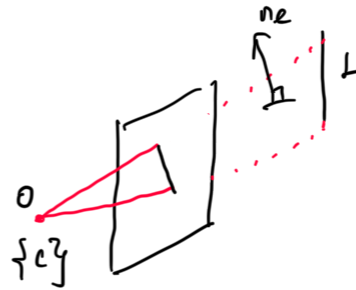
Given line: n_e, v_e, d_e and camera parameters: f_u, f_v, c_u, c_v .

For: $\begin{bmatrix} u_0 \\ v_0 \\ 1 \end{bmatrix} \sim \begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\begin{matrix} \tilde{p} & \underbrace{\quad}_{K} & P_f & & p \sim K P_f \\ \text{pixel on} & & \text{Intrinsic} & & \\ \text{image} & & \text{Matrix} & & \\ & & \text{3D points:} & & \end{matrix}$$

$$P_f^T n_e = 0 \Rightarrow \text{All } P_f \text{ that}$$

$$P_f \sim K^{-1} \tilde{p}$$



$$\text{Hence: } (K^{-1} \tilde{p})^T n_e = 0 \quad \tilde{p}^T K^{-T} n_e = 0 \Rightarrow n_e^T K^{-T} \tilde{p} = 0$$

$$(K^{-T} n_e)^T \tilde{p} = 0$$

$$l = K^{-T} n_e \quad \left\{ \begin{array}{l} \text{2D image line} \\ \tilde{p} \text{ is all the pixel on the line} \end{array} \right.$$

② How to transfer a line from frame $\{C_1\}$ to $\{C_2\}$?

$$L: \{C_1\}: c_{1n_e}, c_{1v_e}, c_{1d_e}$$

$$\{C_2\}: c_{2n_e}, c_{2v_e}, c_{2d_e}$$

Then:

Assume:

